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	For the mobilisation of PBPCs in normal donors prior to allogeneic PBPC transplantation Posology For PBPC mobilisation in normal donors, filgrastim should be administered at 1.0 MU (10 µg/kg/day) for 4 - 5 consecutive days. Leukapheresis should be started at day 5 and continued until day 6 if needed in order to collect a 10⁶ CD34⁺ cells/kg recipient bodyweight. Method of administration Filgrastim should be given by subcutaneous injection. In patients with severe chronic neutropenia (SCN) Posology Constitutional neutropenia: The recommended starting dose is 1.2 MU (12 µg/kg/day) as a single dose or in divided doses. Idiopathic or cyclic neutropenia: The recommended starting dose is 0.5 MU (5 µg/kg/day) as a single dose or in divided doses. Dose adjustments: Filgrastim should be administered daily by subcutaneous injection until the neutrophil count has reached and can be maintained at more than 1.5 x 10⁹/L. When the response has been obtained, the minimal effective dose to maintain this level should be established. Long-term daily administration is required to maintain an adequate neutrophil count. After one to two weeks of therapy, the initial dose may be doubled or halved depending upon the patient's response. Subsequently, the dose may be individually adjusted every 1-2 weeks to maintain the average neutrophil count between 1.5 x 10⁹/L and 10 x 10⁹/L. A faster schedule of dose escalation may be considered in patients presenting with severe infections. In clinical studies, 97% of patients who responded had a complete response at doses of 5 to 24 µg/kg/day. The long-term safety of administration of filgrastim at doses above 24 µg/kg/day in patients with SCN has not been established. Method of administration For congenital, idiopathic or cyclic neutropenia, filgrastim should be given by subcutaneous injection. In patients with HIV infection Posology For reversal of neutropenia: The recommended starting dose of filgrastim is 0.1 MU (1 µg/kg/day), with titration up to a maximum of 0.4 MU (4 µg/kg/day) until a normal neutrophil count is reached and can be maintained (ANC >2.0 x 10⁹/L). In clinical studies, more than 90% of patients responded at these doses, achieving a reversal of neutropenia in a median of 2 days. In a small number of patients (< 10%), doses up to 1.0 MU (10 µg/kg/day) were required to achieve reversal of neutropenia. For maintenance of normal neutrophil counts: When reversal of neutropenia has been achieved, the minimal effective dose to maintain a normal neutrophil count should be established. Initial dose adjustment to alternate day dosing with 30 MU (300 µg/kg) is recommended. Further dose adjustment may be necessary, as determined by the patient's ANC, to maintain the neutrophil count at > 2.0 x 10⁹/L. In clinical studies, dosing with 30 MU (300 µg/kg) on 1-7 days per week was required to maintain the ANC > 2.0 x 10⁹/L, with the median dose frequency being 3 days per week. Long-term administration may be required to maintain the ANC > 2.0 x 10⁹/L. Method of administration For the reversal of neutropenia and maintenance of normal neutrophil counts in patients with HIV infection, filgrastim is administered subcutaneously. Special population Elderly Clinical trials with filgrastim have included a small number of elderly patients but special studies have not been performed in this group and therefore specific posology recommendations cannot be made. Patients with renal impairment Studies that included patients with severe impairment of renal or hepatic function demonstrate that it exhibits a similar pharmacokinetic and pharmacodynamic profile to that seen in normal individuals. Dose adjustment is not required in these circumstances. Paediatric patients in the SCN and cancer settings Sixty-five percent of patients studied in a SCN trial program were under 18 years of age. The safety of the treatment was clear for this age group, which included most patients with congenital neutropenia. There were no differences in the safety profiles for paediatric patients treated for SCN. Data from clinical studies in paediatric patients indicate that the safety and efficacy of filgrastim are similar in both adults and children receiving cytotoxic chemotherapy. The dosage recommendations in paediatric patients are the same as those in adults receiving myelosuppressive cytotoxic chemotherapy. 4.3 Contraindications Hypersensitivity to the active substance or any of the excipients listed in section 6.1. 4.4 Special warnings and precautions for use Accofil is a biosimilar medicinal product of Neupogen. Accofil is not interchangeable or automatically substitutable with Neupogen. Traceability In order to improve the traceability of granulocyte-colony stimulating factors (G-CSFs), the trade name and the batch number of the administered product should be clearly recorded (or stated) in the patient file. Information provided below is based on the innovator data (Neupogen, Amgen Europe B.V.). Special warnings and precautions against indications Filgrastim should not be used to increase the dose of cytotoxic chemotherapy beyond established dosage regimens. Filgrastim should not be administered to patients with severe congenital neutropenia who develop leukapheresis or have evidence of leukemic evolution. Hypersensitivity Hypersensitivity, including anaphylactic reactions, occurring on initial or subsequent treatment have been reported in patients treated with filgrastim. Permanently discontinue filgrastim in patients with clinically significant hypersensitivity. Do not administer filgrastim to patients with a history of hypersensitivity to filgrastim or pegfilgrastim. Immunogenicity As with all therapeutic proteins, there is a potential for immunogenicity. Rates of generation of anti-granulocyte colony stimulating factor (G-CSF) antibodies do occur as expected with all biologics; however, they have not been associated with neutralising activity at present. Special precautions in patients with acute myeloid leukaemia (AML) Malignant cell growth: G-CSF can promote growth of myeloid cells <i>in vitro</i> and similar effects may be seen on some non-myeloid cells <i>in vivo</i>. Myelodysplastic syndrome or Chronic myeloid leukaemia Myelodysplastic syndrome or chronic myeloid leukaemia in patients with myelodysplastic syndrome or chronic myelogenous leukaemia have not been established. Therefore, filgrastim is not indicated for use in these conditions. Particular care should be taken to distinguish the diagnosis of blast transformation of chronic myeloid leukaemia from acute myeloid leukaemia. Acute myeloid leukaemia In studies of limited safety and efficacy data in patients with secondary AML, filgrastim should be administered with caution. The safety and efficacy of filgrastim administration in <i>de novo</i> AML patients aged < 55 years with good cytogenetics [t (8; 21), t (15; 17), and inv (16)] have not been established. Other special precautions Outpockets Monitoring of bone density may be indicated in patients with underlying osteoporotic bone diseases who undergo continuous therapy with filgrastim for more than 6 months. Pulmonary adverse effects Pulmonary adverse reactions, in particular interstitial pneumonia, have been reported after G-CSF administration. Patients with a recent history of pulmonary infiltrates or pneumonia may be at higher risk. The onset of pulmonary signs such as cough, fever and dyspnoea
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	Country	Malaysia		
	Language	Malaysia		CUG
	AW Update Date	25/07/22		
	Prepared By Regulatory Affairs	Checked By Regulatory Affairs		Approved By Quality Assurance

Hepatobiliary disorders		Blood alkaline phosphatase increased Hepatomegaly increased	Gamma-glutamyl transferase increased Aspartate aminotransferase increased		-	-
Skin and subcutaneous tissue disorders	Alopecia Rash Erythema			Sweats syndrome (acute febrile dermatosis) Cutaneous vasculitis	-	-
Musculoskeletal and connective tissue disorders	Musculoskeletal pain	Muscle spasms	Osteoporosis	Bone density decreased Exacerbation of rheumatoid arthritis	-	-
Renal and urinary disorders		Dysuria Haematuria	Proteinuria	Urinary abnormality Glomerulonephritis	-	-
General disorders and administration site conditions	Fatigue Mucosal inflammation Pain	Chest pain Asthenoia Faint Malaise Oedema Peripheral	Injection site reaction		-	-
Injury, poisoning and procedural complications		Transfusion reaction			-	-

* See section 4.8, Description of selected adverse reactions
^ There have been reports of GvHD and fatalities in patients after allogeneic bone marrow transplantation (see section 4.8, Description of selected adverse reactions)
^ Includes bone pain, back pain, arthralgia, myalgia, pain in extremity, musculoskeletal pain, musculoskeletal chest pain, neck pain
^ Cases were observed in the post-marketing setting with filgrastim in patients undergoing bone marrow transplant or PBPC mobilization
^ Adverse events with higher incidence in Filgrastim patients compared to placebo and associated with the sequence of the underlying malignancy or cytotoxic chemotherapy

Description of selected adverse reactions

GvHD
There have been reports of GvHD and fatalities in patients receiving G-CSF after allogeneic bone marrow transplantation (see sections 4.4 and 5.1).

Capillary leak syndrome
Cases of capillary leak syndrome have been reported in the post marketing setting with granulocyte colony-stimulating factor use. These have generally occurred in patients with advanced malignant diseases, sepsis, taking multiple chemotherapy medications or undergoing apheresis (see section 4.4).

In randomised, placebo-controlled clinical studies, filgrastim did not increase the incidence of undesirable effects associated with cytotoxic chemotherapy. In those clinical trials, undesirable effects reported with equal frequency in cancer patients treated with filgrastim/chemotherapy and placebo/chemotherapy included nausea and vomiting, alopecia, diarrhoea, fatigue, anorexia, mucositis, headache, cough, skin rash, chest pain, generalised weakness, sore throat, constipation and pain.

In the post-marketing setting cutaneous vasculitis has been reported in patients treated with filgrastim. The mechanism of vasculitis in patients receiving filgrastim is unknown. The frequency is estimated as uncommon from clinical trial data.

Sweats syndrome
Cases of Sweats syndrome (acute febrile dermatosis) have been reported in the post-marketing setting. The frequency is estimated as uncommon from clinical trial data.

Pulmonary adverse events
In clinical studies and the post-marketing setting pulmonary adverse effects including interstitial lung disease, pulmonary oedema, and lung infiltration have been reported in some cases with an outcome of respiratory failure or acute respiratory distress syndrome (ARDS), which may be fatal (see section 4.4).

Splenomegaly and Splenic rupture
Cases of splenomegaly and splenic rupture have been reported uncommonly following administration of filgrastim. Some cases of splenic rupture were fatal (see section 4.4).

Hypersensitivity
Hypersensitivity-type reactions including anaphylaxis, rash, urticaria, angioedema, dyspnoea and hypotension occurred on or after subsequent treatment in clinical studies and in post-marketing experience. Overall, reports were more common after intravenous administration. In some cases, symptoms have recurred with challenge, suggesting a causal relationship. Filgrastim should be permanently discontinued in patients who experience a serious allergic reaction.

In the post-marketing setting, isolated cases of sickle cell crises have been reported in patients with sickle cell disease (see section 4.4). The frequency is estimated as uncommon from clinical trial data.

Cutaneous vasculitis
Cutaneous vasculitis has been reported in patients treated with Filgrastim. The mechanism of vasculitis in patients receiving Filgrastim is unknown. During long term use cutaneous vasculitis has been reported in 2% of SCN patients.

Pseudogout (chondrocalcinosis pyrophosphate)
Pseudogout has been reported in cancer patients treated with filgrastim, and the frequency is estimated as uncommon from clinical trial data.

Leukocytosis
Leukocytosis (WBC > 50 x 10⁹/L) was observed in 41% of donors and transient thrombocytopenia (platelets < 100 x 10⁹/L) following filgrastim treatment and leukapheresis was observed in 35% of donors.

Paediatric population

Data from clinical studies in paediatric patients indicate that the safety and efficacy of filgrastim are similar in both adults and children receiving cytotoxic chemotherapy suggesting no age-related differences in the pharmacokinetics of filgrastim. The only consistently reported adverse event was musculoskeletal pain which is no different from the experience in the adult population.
There is insufficient data to further evaluate filgrastim use in paediatric subjects.

Other special populations
Geriatric use
No overall differences in safety or effectiveness were observed between subjects over 65 years of age compared to younger adult (>18 years of age) subjects receiving cytotoxic chemotherapy and clinical experience has not identified differences in the responses between elderly and younger adult patients.
There are insufficient data to evaluate Accofil use in geriatric subjects for other approved Accofil indications.

Paediatric SCN patients
Cases of decreased bone density and osteoporosis have been reported in paediatric patients with severe chronic neutropenia receiving chronic treatment with Filgrastim.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdose

The effects of Accofil overdose have not been established. Discontinuation of filgrastim therapy usually results in a 50% decrease in circulating neutrophils within 1 to 2 days, with a return to normal levels in 1 to 7 days.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: immunostimulants, colony stimulating factors, ATC code: L03AA02 Accofil is a bisimil medicinal product.

Pharmacodynamic effects

Information provided below is based on studies conducted with Accofil.
Accofil has been compared to the reference product, Neupogen, in healthy volunteers in four phase 1 studies.
KWI-300-101 was a single-dose, randomized, double-blind, two-way, crossover, comparative study in healthy volunteers to evaluate the pharmacokinetics (PK) and pharmacodynamics (PD) of Accofil and Neupogen. Subjects were randomized to receive single intravenous (IV) dose of 5 microgram (mcg/kg) of either Accofil or Neupogen. The PD results from the per protocol (PP) population are summarized in Table 3. Overall, PD profile of Accofil and Neupogen was found to be similar after a single IV dose of 5 mcg/kg.

Table 3: Pharmacodynamic Parameters of Per Protocol Population Following a Single Intravenous Infusion of 5 mcg/kg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-101)

Endpoint	Accofil (N=35)	Neupogen (N=35)	GM ratio (Accofil/Neupogen)	90% CI (%)
ANC C _{max} (cells x 10 ⁹ /L)	19.02(4.35)	19.28(5.21)	99.5%	93.6 – 105.8
ANCAUC _{0-72h} (min ⁹ /G/L)	46137.4 (8608)	46601.5(9321.6)	-	-

Values presented as mean (SD).
ANC: Absolute neutrophil count; ANC C_{max}: Maximum observed ANC over the sampling interval; AUC_{0-72h}: Area under the curve from 0 hours measured up to 72 hours; CI: Confidence interval; GM: geometric mean

KWI-300-102 was a single-dose, randomized, double-blind, two-way, crossover, comparative study in healthy volunteers to evaluate the PK and PD of Accofil and Neupogen. Subjects were randomized to receive single subcutaneous (SC) dose of 75 mcg and 150 mcg of Accofil or Neupogen. The PD results from the PP population are summarized in Table 4. Overall, PD profile of Accofil and Neupogen was found to be similar after a single SC dose of 75 mcg and 150 mcg.

Table 4: Pharmacodynamic Parameters of Per Protocol Population Following a Single Subcutaneous Injection of 75 and 150 mcg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-102)

Endpoint	Accofil	Neupogen	GM ratio (Accofil/Neupogen)	95% CI (%)
75 mcg				
	N=33	N=33		
ANC C _{max} (G/L)	17.13 (3.74)	18.60 (4.11)	92.0%	87.9 – 96.2
ANC AUC _{0-72h} (min ⁹ /G/L)	35076.8 (6526.3)	37009.8 (7622.5)	-	-
150 mcg				
	N=35	N=35		
ANC C _{max} (G/L)	19.04 (3.83)	19.59 (3.29)	96.3%	91.9 – 101.0
ANC AUC _{0-72h} (min ⁹ /G/L)	43209.3 (7921.5)	43979.6 (6866.4)	-	-

Values presented as mean (SD).
ANC: Absolute neutrophil count; C_{max}: Maximum observed ANC over the sampling interval; AUC_{0-72h}: Area under the curve from 0 hours to 72 hours; CI: Confidence interval; GM: geometric mean

KWI-300-103 was a repeat dose, randomized, double-blind, active and placebo-controlled, parallel group, comparative study to evaluate the PK and PD of Accofil and Neupogen in healthy volunteers. Subjects were randomized to receive daily SC dose of Accofil or Neupogen 5 mcg/kg/day for 4 days, or placebo for 4 days. The PD results from the PP population are summarized in Table 5. Overall, PD profile of Accofil and Neupogen was found to be similar after repeated SC dosing 5 mcg/kg/day.

Table 5: Pharmacodynamic Parameters of Per Protocol Population Following Four Daily Subcutaneous Injections of 5 mcg/kg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-103)

Endpoint	Accofil (N=35)	Neupogen (N=34)	GM ratio (Accofil / Neupogen)	90% CI (%)
ANC C _{max} (G/L) Day 4	30.54 (6.15)	32.27 (7.68)	95.2%	87.29 – 103.85
ANC C _{max-24h} (G/L) Day 1	21.04 (3.68)	21.96 (4.62)	-	-
ANC AUC _{0-24h} (min ⁹ /G/L)	22974.9 (3878.1)	23873.8 (4679.4)	-	-

Values presented as mean (SD).
ANC: Absolute neutrophil count; C_{max}: Maximum observed ANC over the sampling interval; AUC_{0-24h}: area under the curve from 0 hours up to 96 hours; CI: Confidence interval; GM: geometric mean

In KWI-300-103, absolute CD34⁺ cell count was also assessed on study days 1 (i.e., baseline) and 5 (after 4 treatment doses). The mean±SD CD34⁺ cell counts were 2 ± 0.80/μL for Accofil and 1.86 ± 0.75/μL for Neupogen at baseline. Accofil increased CD34⁺ cell counts to 27.65 ± 16.54/μL and Neupogen increased CD34⁺ counts to 24.48 ± 14.98 /μL on day 5.

GCSP-SUIN-05SB01-3FA was a double-blind, single-dose, randomized, three-way, crossover, comparative study in healthy volunteers to evaluate the PK and PD of Accofil, Neupogen-US and Neupogen-EU. Subjects were randomized to receive a single SC dose of 300 mcg Accofil, Neupogen-US or Neupogen-EU. The PD results from the study are summarized in Table 6. Overall, PD profile of Accofil was found to be similar to Neupogen-US and Neupogen-EU after a single SC dose of 300 mcg.

Table 6: Pharmacodynamic Parameters Following a Single Subcutaneous Injection of 300 mcg Accofil, Neupogen-EU or Neupogen-US to Healthy Volunteers (Study GCSP-SUIN-05SB01-3FA)

Endpoint	Accofil (N=43)	Neupogen-US (N=43)	Neupogen-EU (N=44)	GM ratio (Accofil/ Neupogen)	90% CI (%)
ANC C _{max} (cells x 10 ⁹ /L)	20.68	20.49	19.92	1.00 (vs. Neupogen-US); 1.03 (vs. Neupogen-EU)	0.96-1.05 (vs. Neupogen-US); 0.99-1.08 (vs. Neupogen-EU)
ANC AUC ₀₋₄ (h* cells x 10 ⁹ /L)	982.31	964.93	952.13	1.00 (vs. Neupogen-US); 1.03 (vs. Neupogen-EU)	0.97-1.04 (vs. Neupogen-US); 1.00-1.06 (vs. Neupogen-EU)

Values presented as mean.
ANC: Absolute neutrophil count; C_{max}: Maximum observed ANC over the sampling interval; AUC₀₋₄: area under the curve from 0 hours up to sampling time; CI: Confidence interval; GM: geometric mean

Clinical efficacy:
Efficacy of Accofil was evaluated as part of non-comparative, multicenter, repeat-dose study in breast cancer patients receiving TAC (doctaxel, doxorubicin and cyclophosphamide) chemotherapy regimen (KWI-300-104). One-hundred and twenty patients were administered with 5 mcg/kg/day SC dose of Accofil on the second day of chemotherapy cycle and was continued up to 14 days or until post-nadir ANC recovery to normal or near-normal values by laboratory standards; however occurs first. Patients were treated for total 6 cycles. The main efficacy endpoint was duration of severe neutropenia (occurrence of ANC < 0.5 x 10⁹/L) in cycle 1.
The mean (SD) duration of severe neutropenia in cycle was 1.40 (1.07) days. Incidence of severe neutropenia was 77.5% and 2.5%, respectively in cycle 1. No incidence of febrile neutropenia was reported in other cycles. Incidence of severe neutropenia was 3.51%, 7.02%, 4.38%, 7.96% and 10.62%, in cycles 2, 3, 4, 5 and 6 respectively. Frequency of grade 3 and 4 neutropenia in cycle 1 was 68.33% and 77.5% respectively and it was lower in subsequent cycles.

The mean (SD) duration of severe neutropenia in cycle was 1.40 (1.07) days. Incidence of severe neutropenia was 77.5% and 2.5%, respectively in cycle 1. No incidence of febrile neutropenia was reported in other cycles. Incidence of severe neutropenia was 3.51%, 7.02%, 4.38%, 7.96% and 10.62%, in cycles 2, 3, 4, 5 and 6 respectively. Frequency of grade 3 and 4 neutropenia in cycle 1 was 68.33% and 77.5% respectively and it was lower in subsequent cycles.

Information provided below is based on the innovator data (Neupogen, Amgen Europe B.V.)

Human G-CSF is a glycoprotein which regulates the production and release of functional neutrophils from the bone marrow. Accofil containing r-methU-G-CSF (filgrastim) causes marked increases in peripheral blood neutrophil counts within 24 hours, with minor increases in monocytes. In some *severe chronic neutropenia* (SCN) patients, filgrastim can also induce a minor increase in the number of circulating eosinophils and basophils relative to baseline; some of these patients may present with eosinophilia or basophilia already prior to treatment. Elevations of neutrophil counts are dose-dependent at recommended doses. Neutrophils produced in response to filgrastim show normal or enhanced function as demonstrated by tests of chemotactic and phagocytic function. Following termination of filgrastim therapy, circulating neutrophil counts decrease by 50% within 1 to 2 days, and to normal levels within 1 to 7 days.

Use of filgrastim in patients undergoing cytotoxic chemotherapy leads to significant reductions in the incidence, severity and duration of neutropenia and febrile neutropenia. Treatment with filgrastim significantly reduces the duration of febrile neutropenia, antibiotic use and hospitalisation after induction chemotherapy for acute myelogenous leukaemia or myeloblastic therapy followed by bone marrow transplantation. The incidence of fever and documented infections were not reduced in either setting. The duration of fever was not reduced in patients undergoing myeloblastic therapy followed by bone marrow transplantation.

Use of filgrastim, either alone, or after chemotherapy, mobilises haematopoietic progenitor cells into peripheral blood. These autologous peripheral blood progenitor cells (PBPCs) may be harvested and infused after dose-limiting cytotoxic therapy, either in place of, or in addition to bone marrow transplantation. Infusion of PBPCs accelerates haematopoietic recovery, reducing the duration of risk for haemorrhagic complications and the need for platelet transfusions. Recipients of allogeneic PBPCs mobilised with filgrastim experienced significantly more rapid haematological recovery, leading to a significant decrease in time to unsupported platelet recovery when compared with allogeneic bone marrow transplantation. One retrospective European study evaluating the use of G-CSF after allogeneic bone marrow transplantation in patients with acute leukaemia suggests an increase in the risk of graft versus host disease (GvHD), treatment related mortality (TRM) and mortality when G-CSF was administered. In a separate retrospective international study in patients with acute and chronic myelogenous leukaemias, no effect on the risk of GvHD, TRM and mortality was seen. A meta-analysis of allogeneic transplant studies, including the results of nine prospective randomized trials, 8 retrospective studies and 1 case-controlled study, did not detect an effect on the risks of acute GvHD, chronic GvHD or early treatment-related mortality.

Relative risk (95% CI) of GvHD and TRM following treatment with G-CSF after bone marrow(BM) transplantation					
Publication	Period of Study	N	Acute Grade II- IV GvHD	Chronic GvHD	TRM
Meta-Analysis (2003)	1986 - 2001	1198	1.08 (0.87, 1.33)	1.02 (0.82, 1.26)	0.70 (0.38, 1.31)
European Retrospective Study(2004)	1992 - 2002	1789	1.33 (1.08, 1.64)	1.29 (1.02, 1.61)	1.73 (2.32)
International Retrospective Study (2006)	1995 - 2000	2110	1.11 (0.86, 1.42)	1.10 (0.86, 1.39)	1.26 (0.95, 1.67)

Analysis includes studies involving BM transplant during this period; some studies used GM-CSF Analysis includes patients receiving BM transplant during this period

Use of filgrastim for the mobilisation of PBPCs in normal donors prior to allogeneic PBPC transplantation

In normal donors, a 10 micrograms/kg/day dose administered subcutaneously for 4 - 5 consecutive days allows a collection of 4 x 10⁹ CD34⁺ cells/kg recipient body weight in the majority of the donors after two leukaphereses.

Use of filgrastim in adults with SCN (severe congenital, cyclic) and idiopathic neutropenia) induces a sustained increase in absolute neutrophil count (ANC) in peripheral blood and a reduction of infection and related events.

Use of filgrastim in patients with HIV infection maintains normal neutrophil counts to allow scheduled dosing of antiviral and/or other myelosuppressive treatments. There is no evidence that patients with HIV infection treated with filgrastim show an increase in HIV replication.

As with other haematopoietic growth factors, G-CSF has shown *in vitro* stimulating properties on human endothelial cells.

5.2 Pharmacokinetic properties
Information provided below is based on studies conducted with Accofil.
Accofil has been compared to the reference product in healthy volunteers in four phase 1 studies. KWI-300-101 was a single-dose, randomized, double-blind, two-way, crossover, comparative study in healthy volunteers to evaluate the PK and PD of Accofil and Neupogen. Subjects were randomized to receive single IV dose of 5 mcg/kg Accofil or Neupogen. The PK results are summarized in Table 7. Overall, PK profile of Accofil and Neupogen was found to be similar.

Table 7: Pharmacokinetic Parameters of Per Protocol Population Following a Single Intravenous Infusion of 5 mcg/kg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-101)

Endpoint	Accofil (N=51)	Neupogen (N=51)	GM ratio (Accofil/Neupogen)	90% CI (%)
AUC ₀₋₁₂ (min ⁹ ng/mL)	22047494 (4060115)	24340789 (4530366)	90.6%	88.7 - 92.7
C _{max} (pg/mL)	103272.4 (5031.9)	111567 (15688.3)	92.5%	90.3 - 94.7
AUC _{0-24h} (min ⁹ pg/ml)	22075297 (4065640)	24366534 (4536283)	90.7%	88.7 – 92.7
T _{max} (min)	16.3 (9.1)	16.0 (5.5)	-	-

Values presented as mean (SD).
AUC_{0-12h}: Area under the concentration-time curve from 0 hours up to 32 hours; C_{max}: Maximum plasma concentration; T_{max}: Time at which C_{max} is achieved; SD: Standard Deviation; GM: geometric mean

KWI-300-102 was a single-dose, randomized, double-blind, two-way, crossover, comparative study in healthy volunteers to evaluate the PK and PD of Accofil and Neupogen. Subjects were randomized to receive single SC dose of 75 mcg and 150 mcg of Accofil or Neupogen. The PK of Accofil and Neupogen was evaluated in subjects receiving 150 mcg dose. The PK results from the PP population are summarized in Table 8. Overall, PK profile of Accofil and Neupogen was found to be similar after a single SC dose of 150 mcg.

Table 8: Pharmacokinetic Parameters of Per Protocol Population Following a Single Subcutaneous Injection of 150 mcg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-102)

Endpoint	Accofil 150µg (N=51)	Neupogen 150µg (N=55)	GM ratio (Accofil/ Neupogen)	90% CI (%)
AUC ₀₋₁₂ (min ⁹ ng/mL)	3275.7 (919.5)	3414.6 (1093.5)	96.8%	91.0 – 103.0
C _{max} (ng/mL)	7.72 (2.35)	8.35 (3.05)	94.6%	85.9 – 104.1
AUC ₀₋₂₄ (min ⁹ ng/ml)	3282.7 (920.7)	3419.7 (1093.7)	96.8%	91.0 – 103.0
T _{max} (min)	278.6 (41.5)	283.7 (54.1)	-	-

Values presented as mean (SD).
AUC_{0-12h}: Area under the concentration-time curve from 0 hours up to 72 hours; C_{max}: Maximum plasma concentration; T_{max}: Time at which C_{max} is achieved; SD: Standard Deviation; GM: geometric mean

KWI-300-103 was a repeat dose, randomized, double-blind, active and placebo-controlled, parallel group, comparative study in healthy volunteers to evaluate the PK and PD of Accofil and Neupogen. Subjects were randomized to receive daily SC dose of Accofil or Neupogen 5 mcg/kg/day for 4 days, or placebo for 4 days. The PK results from the PP population are summarized in Table 7. Overall, PK profile of Accofil and Neupogen was found to be similar after repeated SC dosing of 5 mcg/kg/day.

Table 9: Pharmacokinetic Parameters of Per Protocol Population Following a Single and Four Daily Subcutaneous Injections of 5 mcg/kg Accofil or Neupogen to Healthy Volunteers (Study KWI-300-103)

Endpoint	Accofil (N=35)	Neupogen (N=34)	GM ratio (Accofil/ Neupogen)	90% CI (%)
AUC _{0-24h} (min ⁹ ng/mL) Day 1	11734.8 (2737)	11839.4 (3392.4)	100.2%	90.3 – 111.1
AUC ₀₋₂₄ (min ⁹ ng/mL) Day 4	5440.8 (1484.6)	5387.6 (1790.2)	102.3%	91.1 – 114.9
C _{max} (ng/mL) Day 1	25.92 (6.95)	25.54 (7.81)	102.2%	91.3 – 114.5
AUC _{0-4hr} (min ⁹ ng/ml) Day 1	11803.8 (2751.5)	11917 (3307.9)	100.1%	90.29 – 110.98
T _{max} (min) Day 1	297.4 (35.2)	307.1 (33.1)	-	-

Values presented as mean (SD).
AUC_{0-24h}: Area under the concentration-time curve from 0 hours up to 24 hours; AUC₀₋₄: Area under the plasma concentration-time curve in steady state, following the last filgrastim administration (trial day 4) up to the last measured filgrastim concentration; C_{max}: Maximum plasma concentration; T_{max}: Time at which C_{max} is achieved; SD: Standard Deviation; CI: Confidence interval; GM: geometric mean.

GCSP-SUIN-05SB01-3FA was a double-blind, single-dose, randomized, three-way, crossover, comparative study in healthy volunteers to evaluate the PK and PD of Accofil, Neupogen-US and Neupogen-EU. Subjects were randomized to receive a single SC dose of 300 mcg Accofil, Neupogen-US or Neupogen-EU. The PK results from the study are summarized in Table 10. Overall, PK profile of Accofil was found to be similar to Neupogen-US and Neupogen-EU after a single SC dose of 300 mcg.

Table 10: Pharmacokinetic Parameters Following a Single Subcutaneous Injection of 300 mcg Accofil, Neupogen-EU or Neupogen-US to Healthy Volunteers (Study GCSP-SUIN-05SB01-3FA)

Endpoint	Accofil (N=43)	Neupogen US (N=43)	Neupogen EU (N=44)	GM ratio (Accofil/ Neupogen)	90% CI (%)
AUC ₀₋₄ (pg*hr/ml)	200720.00	192379.97	186044.48	1.08 (vs. Neupogen-US); 1.10 (vs. Neupogen-EU)	1.02-1.14 (vs. Neupogen-US); 1.04-1.16 (vs. Neupogen-EU)
C _{max} (pg/ml)	24212.80	22456.87	21835.92	1.10 (vs. Neupogen-US); 1.10 (vs. Neupogen-EU)	1.01-1.20 (vs. Neupogen-US); 1.02-1.21 (vs. Neupogen-EU)
AUC _{Inf} (pg*hr/ml)	202126.78	193710.54	187937.67	1.08 (vs. Neupogen-US); 1.09 (vs. Neupogen-EU)	1.02-1.14 (vs. Neupogen-US); 1.04-1.15 (vs. Neupogen-EU)
T _{max} (h)	5.00	5.00	5.00	0.97 (vs. Neupogen-US); 0.99 (vs. Neupogen-EU)	0.88-1.06 (vs. Neupogen-US); 0.91-1.07 (vs. Neupogen-EU)

Values presented as mean.
AUC₀₋₄: Area under the concentration-time curve from 0 hours up to sampling time; AUC_{Inf}: Area under the concentration-time curve from 0 hours up to infinity; C_{max}: Maximum plasma concentration; K_{el}: elimination rate constant; T_{max}: Time at which C_{max} is achieved; SD: Standard Deviation; GM: geometric mean

Information provided below is based on the innovator data (Neupogen, Amgen Europe B.V.)
Absorption

Following subcutaneous administration of recommended doses, serum concentrations were maintained above 10 ng/ml for 8 - 16 hours.

Distribution

The volume of distribution in blood is approximately 150 mL/kg.

Elimination

Clearance of filgrastim has been shown to follow first-order pharmacokinetics after both subcutaneous or intravenous administration. The serum elimination half-life of filgrastim is approximately 3.5 hours, with a clearance rate of approximately 0.6 mL/min/kg. Continuous infusion with Accofil over a period of up to 28 days, in patients recovering from autologous bone-marrow transplantation, resulted in no evidence of drug accumulation and comparable half-lives.

Linearity

There is a positive linear correlation between the dose and the serum concentration of filgr